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Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Heating Effects

English Medium

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HEATING EFFECT

1. HEATING EFFECT OF CARBONATE & BICARBONATE SALTS :

(a) Heating effect of carbonate salts :

Metal carbonate $\xrightarrow{\Delta}$ metal oxide + $CO_2\uparrow$

(i) $MCO_3 \xrightarrow{\Delta} MO + CO_2$ [M = Be, Mg, Ca, Sr, Ba]

(ii) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

(iii) $\underbrace{Cu(OH)_2.CuCO_3}_{\text{Basic Cu(II) carbonate}} \xrightarrow{\Delta} \underset{\text{(black)}}{2CuO} + CO_2 + H_2O$

$\downarrow$ high temperature

$\underset{\text{(red)}}{Cu_2O} + O_2$

(iv) $\underset{\text{white}}{ZnCO_3} \xrightarrow{\Delta} \underset{\substack{\text{Yellow(hot)}\\ \text{white(cold)}}}{ZnO} + CO_2$

(v) $PbCO_3 \xrightarrow{\Delta} \underset{\text{Yellow}}{PbO} + CO_2$

(vi) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

(vii) $(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2$ **Very Important**

(viii) $Pb(OH)_2.2PbCO_3$ (White lead)

- $\xrightarrow{350^oC} Pb_3O_4 + CO\uparrow + CO_2\uparrow + H_2O\uparrow$
- $\xrightarrow[\text{Bunsen burner temperature}]{700-800^oC} 3PbO + 2CO_2\uparrow + H_2O\uparrow$

(ix) All carbonates **except** (Na, K, Rb, Cs) decompose on heating giving CO_2

(x) Carbonates salts of (Na, K, Rb, Cs) do not decompose on heating, they are melt on high temperature.

(xi) Oxides of heavier metals are less stable so further decompose into metal & oxygen

(xii) $\underset{\substack{\text{white}\\ \text{(yellowish white)}}}{Ag_2CO_3} \xrightarrow{\Delta} \underset{\text{Black}}{2Ag} + CO_2 + \frac{1}{2}O_2$

(xi) $HgCO_3 \xrightarrow{\Delta} Hg + \frac{1}{2}O_2\uparrow + CO_2$

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(b) Heating effect of bicarbonate :

(i) Metal bicarbonate $\xrightarrow{\Delta}$ metal carbonate + $CO_2\uparrow$

except (Na, K, Rb, Cs)

$\downarrow$

metal oxide + CO_2

(ii) [General reaction $2HCO_3^- \xrightarrow{\Delta} CO_3^{2-} + H_2O + CO_2$]

(iii) All bicarbonates decompose to give carbonates and CO_2. eg.

(iv) $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

(v) $Mg(HCO_3)_2 \xrightarrow{\Delta} MgO + 2CO_2 + H_2O$

2. HEATING EFFECT OF HYDRATED SULPHATE SALTS :

(i) Metal Sulphate $\xrightarrow{\Delta}$ Metal Oxide + SO_3
(M=Be,Zn,Mg,Ca,Cu,Pb)

SO_3 ($T > 800°C$) $\rightarrow SO_2 + \frac{1}{2}O_2$

CuO ($1100°C$) $\rightarrow \frac{1}{2}Cu_2O + \frac{1}{2}O_2$

(ii) $CuSO_4.5H_2O \xrightarrow[-4H_2O]{100°C} CuSO_4.H_2O \xrightarrow[-H_2O]{220°C} CuSO_4 \xrightarrow{>800°C} CuO + SO_2 + \frac{1}{2}O_2$

Blue vitriol; Bluish White; White

$CuSO_4$ ($750°C$) $\rightarrow CuO + SO_3$ (Black)

(iii) $FeSO_4.7H_2O \xrightarrow[-7H_2O]{300°C} FeSO_4 \xrightarrow[\text{Strongly}]{\Delta} Fe_2O_3 + SO_2 + SO_3$ (very important)

Green vitriol

(iv) $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$

(v) $CaSO_4.2H_2O \xrightarrow{120°C-150°C} (CaSO_4.\frac{1}{2}H_2O) + 1\frac{1}{2}H_2O$

gypsum

(Plaster of paris) or calcium sulphate hemihydrate

$CaSO_4.\frac{1}{2}H_2O$ ($1\frac{1}{2}H_2O$) $\rightarrow CaSO_4.2H_2O$

another crystallographic form of gypsum hard & porous

$CaSO_4.\frac{1}{2}H_2O$ (Δ) $\rightarrow CaSO_4 + \frac{1}{2}H_2O$

Dead burnt

$CaSO_4$ ($1100°C$) $\rightarrow CaO + SO_3$

(vi) $ZnSO_4.7H_2O \xrightarrow[-H_2O]{70°C} ZnSO_4.6H_2O \xrightarrow[-5H_2O]{70°-200°C} ZnSO_4.H_2O \xrightarrow[-H_2O]{>200°C} ZnSO_4 \xrightarrow{\Delta>800°C} ZnO + SO_2 + \frac{1}{2}O_2$

(vii) $MgSO_4.7H_2O \xrightarrow[-7H_2O]{\Delta} MgSO_4\downarrow$ [Same as $ZnSO_4$]

epsom salt

(viii) $2NaHSO_3 \xrightarrow{\Delta} Na_2S_2O_5 + H_2O$

$2NaHSO_3 \xrightarrow{\Delta} Na_2SO_3 + SO_2$

$Na_2SO_3 + SO_2 \rightarrow Na_2S + Na_2SO_4$

(ix) $2NaHSO_4 \xrightarrow{\Delta} Na_2SO_4 + H_2O + SO_3$

(x) $Na_2S_2O_3.5H_2O \xrightarrow{220°C} Na_2S_2O_3 + 5H_2O$

$Na_2S_2O_3$ (Δ, From 4 moles) $\rightarrow 3Na_2SO_4 + Na_2S_5$.

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3. HEATING EFFECT OF NITRATE SALTS

(i) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + $NO_2 + O_2$

(ii) $2M(NO_3)_2 \xrightarrow{\Delta} 2MO + 4NO_2 + O_2$

[M = all bivalent metals eg. Zn^{+2}, Mg^{+2}, Sr^{+2}, Ca^{+2}, Ba^{+2}, Cu^{+2}, Pb^{+2}]

(iii) $2LiNO_3 \xrightarrow{\Delta} Li_2O + 2NO_2 + \frac{1}{2}O_2$

(iv) $MNO_3 \xrightarrow{\Delta} MNO_2 + \frac{1}{2}O_2$

[M=Na,K,Rb,Cs]

↓ high temperature

$M_2O + N_2 + \frac{3}{2}O_2$

(v) $NaNO_3 \xrightarrow{500°C} NaNO_2 + \frac{1}{2}O_2$

(vi) $NaNO_2 \xrightarrow{800°C} Na_2O + N_2 + \frac{3}{2}O_2$

(vii) $BeCl_2 \xrightarrow{N_2O_4} Be(NO_3)_2.2N_2O_4 \xrightarrow[\text{under vaccum}]{\text{Warm to 50°C}} Be(NO_3)_2$ → (700–800°C) $BeO + 2NO_2 + 1/2\ O_2$; → (125°C) $[Be_4O(NO_3)_6]$ basic beryllium nitrate

Exception : If formed oxide is of heavier metal then it being less stable and further decomposed in to metal and oxygen.

(viii) $Hg(NO_3)_2 \xrightarrow{\Delta} Hg + 2NO_2 + O_2$

(ix) $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$

4. HEATING EFFECT OF AMMONIUM SALTS :

If anionic part is oxdising in nature, then N_2 will be the product (some times N_2O).

(i) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$ {This reaction is used for making artificial volcano}

(orange solid) (green)

(ii) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

(iii) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

(iv) $2NH_4ClO_4 \xrightarrow{\Delta} N_2 + Cl_2 + 2O_2 + 4H_2O$

(v) $2NH_4IO_3 \xrightarrow{\Delta} N_2 + I_2 + O_2 + 4H_2O$

If anionic part weakly oxidising or non oxidising in nature then NH_3 will be the product.

(i) $(NH_4)_2HPO_4 \xrightarrow{\Delta} HPO_3 + H_2O + 2NH_3$

(ii) $(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + H_2SO_4$

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(iii) $2(NH_4)_3PO_4 \xrightarrow{\Delta} 2NH_3 + P_2O_5 + 3H_2O$

(iv) $(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2$

(v) $NH_4Cl \xrightarrow{\Delta} NH_3\uparrow + HCl\uparrow$

(vi) $(NH_4)_2S \xrightarrow{\Delta} NH_3\uparrow + H_2S\uparrow$

(vii) $NH_4F \xrightarrow{\Delta} NH_3 + HF$

(viii) $(NH_4)_2MoO_4 \xrightarrow{\Delta} NH_3\uparrow + MoO_3 + H_2O$

5. HEATING EFFECT OF PHOSPHATE SALTS :

(i) 1° Phosphate salts gives metaphosphate salt on heating.

$NaH_2PO_4 \xrightarrow{\Delta} H_2O + NaPO_3$

(ii) 2° Phosphate salts gives pyrophosphate

$Na_2HPO_4 \xrightarrow{\Delta} H_2O + Na_4P_2O_7$

(iii) 3° Phosphate salt have no heating effect

$Na_3PO_4 \xrightarrow{\Delta}$ No effect

(iv) $\underset{\text{microcosmic salt}}{Na(NH_4)HPO_4.4H_2O} \xrightarrow[-4H_2O]{\Delta} NaNH_4HPO_4 \xrightarrow{\text{High temp.}} NaPO_3 + NH_3 + H_2O$

(v) $2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + 2NH_3 + H_2O$

6. HEATING EFFECT OF HALIDES SALTS :

(i) $2FeCl_3 \xrightarrow{\Delta} 2FeCl_2 + Cl_2$

(ii) $AuCl_3 \xrightarrow{\Delta} AuCl + Cl_2$

(iii) $Hg_2Cl_2 \xrightarrow{\Delta} HgCl_2 + Hg$

(iv) $NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$

(v) $Pb(SCN)_4 \xrightarrow{\Delta} Pb(SCN)_2 + (SCN)_2$

(vi) $PbCl_4 \xrightarrow{\Delta} PbCl_2 + Cl_2$

(vii) $PbBr_4 \xrightarrow{\Delta} PbBr_2 + Br_2$ **[PbI_4 does not exists]**

(viii) $\underset{\text{scarlet red}}{HgI_2} \underset{\text{On Rubbing}}{\overset{127°C}{\rightleftharpoons}} \underset{\text{yellow}}{HgI_2}$

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7. HEATING EFFECT OF HYDRATED CHLORIDE SALTS

(i) $MgCl_2 . 6H_2O \xrightarrow{\Delta} MgO + 2HCl + 5H_2O$

(ii) $2FeCl_3 . 6H_2O \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$

(iii) $2AlCl_3 . 6H_2O \xrightarrow{\Delta} Al_2O_3 + 6HCl + 9H_2O$

(iv) $\underset{\text{Pink}}{CoCl_2.6H_2O} \xrightarrow[-2H_2O]{50^\circ C} CoCl_2.4H_2O \xrightarrow[-2H_2O]{58^\circ C} \underset{\text{Red violet}}{CoCl_2.2H_2O} \xrightarrow[-2H_2O]{140^\circ C} \underset{\text{Blue}}{CoCl_2}$

Hydrated Co^{2+} salt - Pink Anhydrous Co^{2+} salt - Blue

8. HEATING EFFECT OF OXIDE :

(i) $2Ag_2O \xrightarrow[300^\circ C]{\Delta} 4Ag + O_2$

(ii) $\underset{\text{white}}{ZnO} \underset{\text{Cold}}{\overset{\text{Hot}}{\rightleftharpoons}} \underset{\text{yellow}}{ZnO}$

(iii) $PbO_2 \xrightarrow{\Delta} PbO + \frac{1}{2}O_2$

(iv) $\underset{\text{yellow}}{PbO}$ (Massicot) $\xrightarrow[\text{Cooled \& powdered}]{\text{Fused}} \underset{\text{Litharge (red)}}{PbO}$

(v) $3MnO_2 \xrightarrow{900^\circ C} Mn_3O_4 + O_2$

(vi) $\underset{\text{Red lead}}{Pb_3O_4} \underset{350^\circ C}{\overset{500^\circ C}{\rightleftharpoons}} 6PbO + O_2$ (Litharge)

(vii) $2CrO_5 \xrightarrow{\Delta} Cr_2O_3 + \frac{7}{2}O_2$

(viii) $\underset{\text{(White)}}{K_2O} \underset{\text{cold}}{\overset{\Delta}{\rightleftharpoons}} \underset{\text{(yellow)}}{K_2O}$

(ix) $I_2O_5 \xrightarrow{\Delta} I_2 + \frac{5}{2}O_2$

(x) $\underset{\text{yellow}}{HgO} \xrightarrow{\Delta} \underset{\text{red}}{HgO} \xrightarrow[\text{strong heating}]{400^\circ C} Hg + \frac{1}{2}O_2$

(xi) $2CrO_3 \xrightarrow{420^\circ C} Cr_2O_3 + \frac{3}{2}O_2$

9. HEATING EFFECT OF PERMANGANATE :

$\underset{\text{dark puple (almost black)}}{2KMnO_4} \xrightarrow{513K} \underset{\text{(green)}}{K_2MnO_4} + \underset{\text{(black)}}{MnO_2} + O_2$

10. HEATING EFFECT OF DICHROMATE & CHROMATE SALTS :

$\underset{\text{orange}}{2K_2Cr_2O_7} \xrightarrow{\Delta} \underset{\text{yellow}}{2K_2CrO_4} + \underset{\text{green}}{Cr_2O_3} + \frac{3}{2}O_2\uparrow$

11. HEATING EFFECT OF ACIDS :

(i) $\underset{\text{conc.}}{2HNO_3} \xrightarrow{\Delta} H_2O + 2NO_2 + \frac{1}{2}O_2$

(ii) $H_2SO_4 \xrightarrow{444^\circ C} H_2O + SO_3$

(iii) $H_2SO_4 \xrightarrow{>800^\circ C} H_2O + SO_2 + \frac{1}{2}O_2$

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(iv) $3H_2SO_3 \longrightarrow 2H_2SO_4 + S\downarrow + H_2O$

(v) $3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

(vi) $HClO_3 \longrightarrow HClO_4 + ClO_2 + H_2O$

(vii) $3HOCl \longrightarrow 2HCl + HClO_3$

(viii) $4H_3PO_3 \xrightarrow[200°C]{\Delta} 3H_3PO_4 + PH_3$

(ix) $2H_3PO_2 \longrightarrow H_3PO_4 + PH_3$

(x) $2NaH_2PO_2 \longrightarrow Na_2HPO_4 + PH_3$

} Undergoes disproportionation reaction

(xi) $H_2C_2O_4 \xrightarrow{\Delta} H_2O + CO + CO_2$

(xii) $H_3PO_4 \xrightarrow{220°C} H_4P_2O_7 \xrightarrow{320°C} 4HPO_3 \xrightarrow{>320°C} 2P_2O_5 + 2H_2O$

(xiii) $H_3BO_3 \xrightarrow{100°C} 4HBO_2 \xrightarrow{140°C} H_2B_4O_7 \xrightarrow[hot]{Red} 2B_2O_3 + H_2O$

12. HEATING EFFECTS OF ACETATE SALTS

Metal acetate $\xrightarrow{\Delta}$ Metal Carbonate (M = LiBe Mg) + CH_3COCH_3

Metal Carbonate $\xrightarrow{\Delta}$ Metal oxide + CO_2

(i) $Pb(OAc)_2 \xrightarrow{\Delta} PbO + CO_2 + CH_3COCH_3$

(ii) $Mg(OAc)_2 \xrightarrow{\Delta} MgO + CO_2 + CH_3COCH_3$

(iii) $Be(OAc)_2 \xrightarrow{\Delta} BeO + CO_2 + CH_3COCH_3$

(iv) $Ca(OAc)_2 \xrightarrow{\Delta} CaCO_3 + CH_3COCH_3$

(v) $Ba(OAc)_2 \xrightarrow{\Delta} BaCO_3 + CH_3COCH_3$

(vi) $CH_3CO_2K \xrightarrow{\Delta} K_2CO_3 + CH_3COCH_3$

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13. HEATING EFFECTS OF OXALATE SALTS

$$\text{Metal oxalate} \xrightarrow{\Delta} \underset{\substack{\text{Carbonate}\\(M = LiBe\ Mg)}}{\text{Metal}} + CO + CO_2$$

$$\text{Metal carbonate} \xrightarrow{\Delta} \text{Metal oxide} + CO_2$$

(i) $\boxed{7Na_2C_2O_4 \xrightarrow{\Delta} 7Na_2CO_3 + 2CO_2 + 3CO + 2C}$

(ii) $SnC_2O_4 \xrightarrow{\Delta} SnO + CO_2 + CO$

(iii) $FeC_2O_4 \xrightarrow{\Delta} FeO + CO + CO_2$

(iv) $Ag_2C_2O_4 \xrightarrow{\Delta} 2Ag + 2CO_2$

(v) $HgC_2O_4 \xrightarrow{\Delta} Hg + 2CO_2$

14. HEATING EFFECTS OF FORMATE SALTS

(i) $HCO_2Na \xrightarrow{350°C} Na_2C_2O_4 + H_2\uparrow$

(ii) $HCOOAg \xrightarrow{\Delta} HCOOH + 2Ag + \underbrace{\frac{1}{2}O_2 + CO}_{\downarrow\ CO_2}$

(iii) $(HCOO)_2Hg \longrightarrow HCOOH + Hg + \underbrace{\frac{1}{2}O_2 + CO}_{\downarrow\ CO_2}$

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